Regression and Latent Variable Models for Bounded Outcomes

An Introduction Through the Chapter Examples

Jorge Luis Bazán, University of São Paulo

October 20, 2026

1. Introduction: Motivation and Scope

- Bounded outcomes have known lower and upper limits.
- Examples include proportions, scores out of twenty, line-segment ratings, and municipal indicators.
- The limits help determine the response distribution and interpretation.
- The scientific objective determines whether to use regression, continuous latent traits, or discrete attribute profiles.

1.1 Four Chapters and Their Questions

1. **Continuous regression:** How does a proportion's distribution change with observed covariates?
2. **Bounded counts:** How can a model accommodate variability and tails beyond the binomial?
3. **Bounded IRT:** What do bounded item ratings reveal about a continuous latent trait?
4. **Bounded CDMs:** Which combinations of latent attributes are supported by continuous indicators?

1.2 Motivating Paper: Continuous Regression

Piccirilli et al. (2026)

Flexible bounded mixed regression models for studies of longitudinal cancer mortality rate in Brazil

- Flexible distributions for continuous bounded responses.
- Mixed models address longitudinal dependence.
- Brazilian cancer mortality provides the application motivation.
- Chapter 1 introduces estimation through independent simulations.

1.3 Motivating Paper: Bounded Counts

Bayes et al. (2024)

A robust regression model for bounded count health data

- The beta-2-binomial (B2B) extends the beta-binomial (BB), adding shape flexibility at fixed mean and variance.
- The known upper count remains part of the response model.
- Chapter 2 combines simulations with inappropriate hospital days.
- Separate submodels describe mean, dispersion, and shape.

1.4 Motivating Paper: Bounded IRT

Molenaar et al. (2022)

Zero and one inflated item response theory models for bounded continuous data

- Continuous item scores measure a latent person trait.
- Endpoint probabilities complement the interior density.
- Chapter 3 uses Abasement ratings to introduce item and person parameters.
- Bayesian estimation quantifies measurement uncertainty.

1.5 Motivating Paper: Bounded Cognitive Diagnosis

Oliveira et al. (2026)

A cognitive diagnosis model for latent classification of bounded continuous variables

- B-DINA classifies discrete latent attribute profiles.
- Continuous bounded indicators retain their observed scale.
- Chapter 4 links municipal indicators to Economy, Health, and Education through a Q-matrix.
- Posterior profile probabilities express classification uncertainty.

2. Measurement Scale: Continuous Outcomes

For a measurement Y' with theoretical bounds $L < U$,

$$Y = \frac{Y' - L}{U - L}.$$

- Rescaling places responses on the unit interval.
- Bounds come from the measurement definition.
- Rescaling preserves order; it does not make discrete scores continuous.
- An ordinary beta density models $0 < Y < 1$ and assigns no mass to either endpoint.
- Exact zeros and ones require explicit consideration of endpoint probabilities.

2.1 Measurement Scale: Bounded Counts

$$C \in \{0, 1, \ldots, m\}, \qquad m \text{ known.}$$

- Dividing by m leaves a finite grid of possible responses.
- $1/2$ and $10/20$ have equal proportions but different binomial sampling variances.
- The denominator carries information about the response mechanism.
- A known upper limit alone does not establish a binomial model.

3. Continuous Regression: Motivating Examples

- **Application motivation:** longitudinal cancer mortality rates in Brazil (Piccirilli et al. 2026).
- **Chapter practice:** independent simulated responses with an observed covariate.
- One scenario uses a beta conditional distribution.
- Another uses a logistic skew-normal distribution from the Johnson-S_B type (JSBt) construction.
- Simulations allow comparison with known generating mechanisms.

3.1 Continuous Regression: Parameter Interpretation

Both simulation scenarios use the predictor

$$\eta_i = -0.40 + 1.20 x_i.$$

- **Beta:** $\mu_i = \text{logit}^{-1}(\eta_i)$ is the conditional mean (Ferrari and Cribari-Neto 2004).
- **JSBt:** η_i controls baseline location on the logit-transformed scale (Rodrigues et al. 2020).
- Its inverse-logit is a response-scale location index, with a meaning that depends on the family.
- Parameter interpretation must follow the chosen distribution.

3.2 Continuous Regression: Estimation and Checking

- GAMLSS allows covariates in location, scale, and available shape parameters (Rigby and Stasinopoulos 2005).
- AIC and BIC compare penalized fit across candidate distributions.
- Quantile residuals assess remaining distributional discrepancies (Dunn and Smyth 1996).
- The chapter fits independent responses in its simulations.
- The motivating mixed models also address longitudinal or clustered dependence (Piccirilli et al. 2026).

4. Bounded Counts: The Binomial Starting Point

The first simulation uses a maximum of twenty:

$$C_i \mid x_i \sim \text{Binomial}(20, \mu_i), \qquad \text{logit}(\mu_i) = -0.4 + 1.1 x_i.$$

$$E(C_i \mid x_i) = 20\mu_i,$$
$$\text{Var}(C_i \mid x_i) = 20\mu_i(1 - \mu_i).$$

The mean fixes the variance under the binomial model.

4.1 Bounded Counts: Dispersion and Shape

For BB and B2B, conditional on covariates,

$$E(C_i) = m_i \mu_i,$$
$$\text{Var}(C_i) = m_i \mu_i (1 - \mu_i)\{1 + (m_i - 1)\delta_i\}.$$

- m_i: known upper count for observation i.
- δ_i: extra-binomial variation.
- B2B adds ν_i to modify shape while preserving this mean and variance at fixed μ_i and δ_i (Bayes et al. 2024).
- Simulations compare binomial, BB, and B2B fits.

4.2 Bounded Counts: Inappropriate Hospital Days

The application uses `aep` from `gamlss.data` (Bayes et al. 2024).

- Response: inappropriate days spent in hospital.
- The repository reports a B2B fit to 1,383 observations.
- **Mean:** `ward`, `loglos`, and `year`.
- **Dispersion:** `year` and `ward`.
- **Shape:** `year`.
- Separate submodels describe expected counts, extra-binomial variation, and distributional shape.

4.3 Bounded Counts: Application Interpretation

Reported mean-submodel associations, holding other mean predictors fixed:

- Wards 2 and 3 have lower expected counts than the baseline ward.
- `loglos` and `year90` are associated with higher expected counts.
- Dispersion and shape coefficients describe separate distributional features.
- Coefficients lie on link scales. Inverse links give the corresponding parameter values.
- Boundary checks compare observed zero and maximum counts with fitted probabilities.

These are model-based associations.

4.4 Bounded Counts: Practical Estimation

- Using GAMLSS and `RobustDiscrete`, fit BB first, then initialize B2B from its fitted values.
- Convert the GAMLSS BB scale to the B2B dispersion start:

$$\delta_{\text{start}} = \frac{\sigma_{\text{BB}}}{1 + \sigma_{\text{BB}}}.$$

- B2B likelihood evaluation requires numerical integration.
- Compare AIC/BIC, normalized randomized quantile residuals (Dunn and Smyth 1996), and convergence.
- The chapter displays application code and reported results without running that fit during rendering.

5. Bounded IRT: The Abasement Example

- **Data:** 244 people respond to 10 adjectives.
- Ratings use a 60-millimeter line segment, rescaled to $[0, 1]$.
- Higher responses, after reverse scoring, indicate greater endorsement of self-critical attitudes and inadequacy.
- Of 2,440 responses, approximately 10.74% are exact zeros, with no exact ones.
- The measurement target is a continuous latent person trait (Molenaar et al. 2022).

5.1 Bounded IRT: Item and Person Parameters

For person i and item j,

$$\eta_{ij} = a_j \theta_i + b_j.$$

- θ_i: continuous latent person position.
- $a_j > 0$: item discrimination.
- b_j: item easiness under this sign convention.
- Candidate interior distributions include Johnson-S_B, beta, and simplex.
- `BoundIRT` supports Bayesian estimation and interpretation of item and person estimates.

5.2 Bounded IRT: Endpoint Responses

A general zero and one inflated model assigns

$$\Pr(Y_{ij} = 0) = \pi_{0,ij}, \qquad \Pr(Y_{ij} = 1) = \pi_{1,ij},$$

with remaining weight

$$1 - \pi_{0,ij} - \pi_{1,ij}$$

on an interior response density (Molenaar et al. 2022).

- The Abasement example motivates zero inflation.
- Predictive checks assess endpoint frequencies and interior responses.
- Person estimates include uncertainty about the latent trait.

6. Bounded CDMs: Municipal Indicators

- **Data:** 19 continuous indicators for 1,668 municipalities in Southeastern Brazil.
- Latent domains are **Economy, Health, and Education** (Oliveira et al. 2026).
- Coding makes larger responses represent more favorable conditions.
- For example, income equality uses $1 - \mathrm{GINI}$.
- A Q-matrix specifies which domains each indicator measures.

The chapter uses the supplied data's 1,668 rows and documents a published introductory count of 1,688.

6.1 Bounded CDMs: Attribute Profiles

$$\boldsymbol{\alpha}_i = (\alpha_{i,\text{Economy}}, \alpha_{i,\text{Health}}, \alpha_{i,\text{Education}}) \in \{0,1\}^3.$$

- Three binary attributes yield eight possible profiles.
- $(1,0,1)$ represents favorable Economy and Education with unfavorable Health under the model.
- Posterior profile probabilities express classification uncertainty.
- The labels describe inferred conditions, without externally validated policy thresholds.
- The Q-matrix specifies measurement relationships and does not establish causal effects.

6.2 Bounded CDMs: The B-DINA Rule

The conjunctive gate for an indicator's required attributes is

$$\xi_{ij} = \prod_{k:q_{jk}=1} \alpha_{ik}.$$

- An indicator requiring Economy and Health enters its higher-mean regime only when both are present.
- Two beta response regimes can differ in mean and precision.
- Indicators remain continuous, while latent attributes are discrete.
- Bayesian estimation in JAGS yields profile probabilities and item parameters.
- MCMC diagnostics and predictive checks support interpretation.

7. Regression, IRT, and Cognitive Diagnosis

Approach	Explanatory structure	Main output
Bounded regression	Observed covariates	Conditional response distribution
Bounded IRT	Continuous latent trait	Person positions and item parameters
Bounded CDM	Discrete latent attributes and Q-matrix	Posterior profile probabilities

The choice follows the research objective and the measurement structure.

7.1 A Shared Analysis Workflow

1. **Describe observations:** theoretical bounds, support, endpoints, and dependence.
2. **State the target:** covariate effects, latent trait, or attribute profiles.
3. **Choose the conditional model:** response family and parameter interpretation.
4. **Estimate and check:** computational diagnostics and predictive fit.
5. **Communicate results:** relevant response or latent scale, with uncertainty.

7.2 Practical Tools and Course Outcomes

- **GAMLSS:** continuous and bounded count regression (Stasinopoulos and Rigby 2007).
- **`BoundIRT`:** bounded continuous item response models.
- **JAGS through `rjags`:** Bayesian bounded cognitive diagnosis.
- The chapter examples develop interpretation alongside estimation.
- Further questions concern flexible distributions, dependence, computation, and diagnostic uncertainty.

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